Diversity

1.20 The reef systems in the Coral Sea are dominated by spectacular sponge gardens, and support high biodiversity. Approximately 745 species of molluscs (shellfish, squid and octopus) have been found, including several that are considered rare. Six hundred and

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SOURCE: MEPC.268(68).pdf
and support high biodiversity. Approximately 745 species of molluscs (shellfish, squid and octopus) have been found, including several that are considered rare. Six hundred and twenty eight species of fish are known to occur in the Coral Sea. The small islets and cays of the Coral Sea are important nesting places for many species of seabirds.

1.21 Reefs in the area provide a habitat mosaic for diverse and abundant invertebrate and fish communities. A diversity of hard and soft corals, sponge gardens, crustaceans and molluscs are found in the area, as well as a distinct Coral Sea reef fish community that includes many unique species.

1.22 Significant variation in water depth and sea floor features are contributing factors to the high levels of species diversity in the area. Sections of the continental shelf have a mosaic of rocky reefs and soft sediments and support species from a diverse range of taxonomic groups. The extensive seamount systems of the Coral Sea contain a large variety of sponges, corals, gorgonians, sea squirts and crinoids, the latter of which can grow unusually large and are frequently very long-lived, often exceeding several hundred years.

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of sponges, corals, gorgonians, sea squirts and crinoids, the latter of which can grow unusually large and are frequently very long-lived, often exceeding several hundred years.

1.23 Abyssal regions are yet to be fully explored, but there is evidence of biologically important systems likely to contain a vast reservoir of undiscovered species.

Productivity

1.24 The pelagic environment of the area is akin to a vast desert with small oases of biodiversity and productivity. These areas of primary productivity influence the spread of algae, one of the area's most abundant and diverse life forms, covering a greater region than corals and forming an important part of the food chain. Areas with a high biomass of algae

increase planktonic activity and create high levels of prey abundance, attracting aggregations

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increase planktonic activity and create high levels of prey abundance, attracting aggregations of higher order herbivorous and apex predator species. These localized productivity hot spots in an otherwise nutrient-poor environment provide habitat, migration and dispersal corridors for many iconic and endangered species.

1.25 Marine species and seabirds can journey hundreds or even thousands of kilometres to breed in the Coral Sea, or to travel through en route to breeding areas beyond the region. Areas of high productivity such as the seamounts are therefore critical "stepping stones" within the barren open ocean and are important aggregators for a range of species including lantern fish, albacore tuna, billfish and sharks. These species rely on foraging opportunities supplied by productivity hot spots in the Coral Sea to sustain them on their journey. Large marine mammals journey many kilometres to breed in the Reserve, or to travel through en route to breeding areas.

1.26 It is also thought that the organic particulates contained in nutri ent-rich intrusions in

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marine mammals journey many kilometres to breed in the Reserve, or to travel through en route to breeding areas.

1.26 It is also thought that the organic particulates contained in nutri ent-rich intrusions in the area are responsible for the settlement and sustained growth of coral reef ecosystems, which have the highest gross primary productivity of all ocean ecosystems. These waters carry organic matter into the region where they contribute significantly to the overall productivity of the system.

Spawning and breeding grounds

1.27 The area contains critical habitat features used by numerous species to spawn and breed and which are therefore essential for their survival.

.1 The waters over the Queensland and Townsville Troughs appear important for attracting aggregations of large pelagic species, either to feed or spawn.

.2 Extensive seabird rookeries within the small islets and cays of the area are of global and national importance.

.3 T he area is a major feeding and breeding location for six of the world's seven species of sea turtles, all of which are listed on the IUCN Red List of Threatened Species.

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seven species of sea turtles, all of which are listed on the IUCN Red List of Threatened Species.

.4 Particularly sheltered regions, such as lagoons, are thought to be important nursery sites for sharks and predatory fish, while other species, such as the dwarf minke whale, the hawksbill and leatherback turtles and the endangered Herald petrel, forage in these calmer areas.

.5 The northern extension of the Tasmantid Seam ount Chain, where the seamounts extend to the surface and are capped by islands and reefs, provide feeding and breeding grounds for open ocean species including billfish, marine turtles and marine mammals.

.6 Each year from September to December, black marlin aggregate in the area to spawn. This is the only known spawning location for black marlin in the world.

Naturalness

1.28 The Coral Sea is considered one of the most distinctive and undisturbed natural systems in the world.

RESOLUTION MEPC.268(68)
(adopted on 15 May 2015)
DESIGNATION OF THE SOUTH-WEST CORAL SEA AS AN EXTENSION OF THE GREAT BARRIER REEF AND

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Annex 14, page 8

https://edocs.imo.org/Final Documents/English/MEPC 68-21-ADD.1 (E).doc
1.29 The area is not directly threatened by land-based sources of pollution and has relatively low levels of fishing. It is one of the world's last tropical oceanic regions containing high biodiversity coral reefs that are virtually pristine, and where large populations of pelagic predators have not been severely depleted. The topography of the area has also contributed to its pristine nature, with recent surveys identifying deep-sea ecosystems which have remained largely unchanged for millions of years.
1.30 The Coringa Cays and Lihou Islets contain important bird and turtle nesting sites that are almost totally free from anthropogenic disturbances such as lighting, beach use, pollution, feral animals, and boat traffic compared to nesting sites throughout the GBR.

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that are almost totally free from anthropogenic disturbances such as lighting, beach use,

pollution, feral animals, and boat traffic compared to nesting sites throughout the GBR. These areas, therefore, are reference sites to determine the impacts of such disturbances on breeding success within the populations.

Integrity

1.31 The extension area covers a large area, and encompasses parts of six provincial bioregions identified in the Integrated Marine and Coastal Regionalisation of Australia Version 4.0 (IMCRA v.4.0). These bioregions contain a wide variety of interconnected habitats, and also provide important "stepping-stone" links between the GBR and the wider Pacific Ocean.

1.32 The integrity of the area is in part due to its remote nature, with the nearest point to a mainland coast over 60 kilometres away, and the furthest point 1100 kilometres.

1.33 The area therefore demonstrates the characteristics of an isolated, effectively self-sustaining ecological unit, as evidenced by the high proportion of endemism both within the region, and between individual reef communities.

Fragility

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